PART II: PATTERNS

Topic 11: Outliers and Spikes

The Core Concept

An **outlier** is a data point so far outside the normal range that it likely reflects a **specific, one-off event** rather than a structural shift in the underlying pattern. Unlike noise (which is small), a spike is massive and typically traceable to a single cause.

Why It Matters

Outliers can completely distort averages and mislead long-term analysis. Including them without flagging them gives a false picture of the underlying situation.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

In October 2023, Precinct South recorded 85 use-of-force incidents — more than three times the typical monthly average of 25. This was caused by a week of major civil unrest.

With the outlier included: Annual average for South = ~31. Looks like a worsening trend.

With the outlier identified and noted: Annual average for South = ~25 (excluding October). The precinct is actually stable.

Neither approach is dishonest — but they tell very different stories. The correct approach is to report both, with clear explanation.